

An Analysis of Image segmentation

Image segmentation

Image segmentation -- Part 1

Multi-scale segmentation Image segmentations are computed at multiple scales in scale space and sometimes propagated from coarse to fine scales; see scale-space segmentation. Segmentation criteria can be arbitrarily complex and may take into account global as well as local criteria. A common requirement is that each region must be connected in some sense.

Image segmentation -- Part 2

This mathematical expression can be implemented by convolving with an appropriate mask. If we extend this equation to three dimensions (x,y,z) , the intensity at each pixel location around a central pixel at (x, y, z) is replaced by their corresponding values. This equation becomes particularly useful when we assume that all pixels have unit spacing along each axis. A sphere mask has been developed for use with three-dimensional datasets. The sphere mask is designed to use only integer arithmetic during calculations, thereby eliminating the need for floating-point hardware or software. When applying these concepts to actual images represented as arrays of numbers, we need to consider what happens when we reach an edge or border region. The function $g(x, y)$ is defined as:

Image segmentation -- Part 3

Edge detection Edge detection is a well-developed field on its own within image processing. Region boundaries and edges are closely related, since there is often a sharp adjustment in intensity at the region boundaries. Edge detection techniques have therefore been used as the base of another segmentation technique. The edges identified by edge detection are often disconnected. To segment an object from an image however, one needs closed region boundaries. The desired edges are the boundaries between such objects or spatial-taxons. Spatial-taxons are information granules, consisting of a crisp pixel region, stationed at abstraction levels within a hierarchical nested scene architecture. They are similar to the Gestalt psychological designation of figure-ground, but are extended to include foreground, object groups, objects and salient object parts. Edge detection methods can be applied to the spatial-taxon region, in the same manner they would be applied to a silhouette. This method is particularly useful when the disconnected edge is part of an illusory contour. Segmentation methods can also be applied to edges obtained from edge detectors. Lindeberg and Li developed an integrated method that segments edges into straight and curved edge segments for parts-based object recognition, based on a minimum description length (MDL) criterion that was optimized by a split-and-merge-like method with candidate breakpoints obtained from complementary junction cues to

obtain more likely points at which to consider partitions into different segments.

Image segmentation -- Part 4

Simulated annealing (SA) Derived as an analogue of annealing in metallurgy, simulated annealing (SA) uses change in pixel label over iterations and estimates the difference in energy of each newly formed graph to the initial data. If the newly formed graph is more profitable, in terms of low energy cost, given by:

Image segmentation -- Part 5

$$\operatorname{argmin}_{u,K} \gamma |K| + \mu \int K C |\nabla u|^2 dx + \int (u-f)^2 dx.$$

Image segmentation -- Part 6

Watershed transformation The watershed transformation considers the gradient magnitude of an image as a topographic surface. Pixels having the highest gradient magnitude intensities (GMIs) correspond to watershed lines, which represent the region boundaries. Water placed on any pixel enclosed by a common watershed line flows downhill to a common local intensity minimum (LIM). Pixels draining to a common minimum form a catch basin, which represents a segment.

Image segmentation -- Part 7

the algorithm selects the newly formed graph. Simulated annealing requires the input of temperature schedules which directly affects the speed of convergence of the system, as well as energy threshold for minimization to occur.

Image segmentation -- Part 8

Content-based image retrieval Machine vision Medical imaging, and imaging studies in biomedical research, including volume rendered images from computed tomography, magnetic resonance imaging, as well as volume electron microscopy techniques such as FIB-SEM. Locate tumors and other pathologies Measure tissue volumes Diagnosis, study of anatomical structure Surgery planning Virtual surgery simulation Intra-surgery navigation Radiotherapy Digital Pathology and Histopathology. Nuclei instance segmentation in variously stained whole slide images (WSIs) refers to the automatic delineation of individual cells' nuclei borders. This task is a specialized subproblem of instance segmentation, with high practical importance in biomedical research and diagnostics (e.g. cell counting, morphometric feature extraction, tumor grading, prognostic biomarker extraction). The modern approach to the

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problem is the use of deep learning architectures developed to encompass the main challenges of separating overlapping or fused nuclei. The availability of few humanly-annotated datasets poses another challenge to the problem. Object detection Pedestrian detection Face detection Brake light detection Locate objects in satellite images (roads, forests, crops, etc.) Recognition Tasks Face recognition Fingerprint recognition Iris recognition Prohibited Item at Airport security checkpoints Traffic control systems Video surveillance Video object co-segmentation and action localization Several general-purpose algorithms and techniques have been developed for image segmentation. To be useful, these techniques must typically be combined with a domain's specific knowledge in order to effectively solve the domain's segmentation problems.

Image segmentation -- Part 9

Markov random fields The application of Markov random fields (MRF) for images was suggested in early 1984 by Geman and Geman. Their strong mathematical foundation and ability to provide a global optimum even when defined on local features proved to be the foundation for novel research in the domain of image analysis, de-noising and segmentation. MRFs are completely characterized by their prior probability distributions, marginal probability distributions, cliques, smoothing constraint as well as criterion for updating values. The criterion for image segmentation using MRFs is restated as finding the labelling scheme which has maximum probability for a given set of features. The broad categories of image segmentation using MRFs are supervised and unsupervised segmentation.

Image segmentation -- Part 10

The Laplacian operator is employed such that the partial derivatives are derived from a specific equation. The second partial derivative of $f(x, y)$ with respect to x and y are given by:

Image segmentation -- Part 11

Pick K cluster centers, either randomly or based on some heuristic method, for example K-means++ Assign each pixel in the image to the cluster that minimizes the distance between the pixel and the cluster center Re-compute the cluster centers by averaging all of the pixels in the cluster Repeat steps 2 and 3 until convergence is attained (i.e. no pixels change clusters) In this case, distance is the squared or absolute difference between a pixel and a cluster center. The difference is typically based on pixel color, intensity, texture, and location, or a weighted combination of these factors. K can be selected manually, randomly, or by a heuristic. This algorithm is guaranteed to

converge, but it may not return the optimal solution. The quality of the solution depends on the initial set of clusters and the value of K . The Mean Shift algorithm is a technique that is used to partition an image into an unknown apriori number of clusters. This has the advantage of not having to start with an initial guess of such parameter which makes it a better general solution for more diverse cases.